



Numerical simulation of vortex flow field generated in a novel nested-bottled photobioreactor to improve *Arthrospira platensis* growth

Ameer Ali Kubar^a, Santosh Kumar^a, Wei Liu^d, Yi Cui^a, Feifei Zhu^b, Xiangru Xu^c, Cong Shao^a, Xinjuan Hu^a, Nama Yaa Akyea Prempeh^a, Shuhao Huo^{a,*}, Conceptualization, Resources, Writing-review & editing, Supervision, Project administration.

^a School of Food and Biological Engineering, Jiangsu University, Zhenjiang 212013, China

^b School of Life Sciences, Jiangsu University, Zhenjiang 212013, China

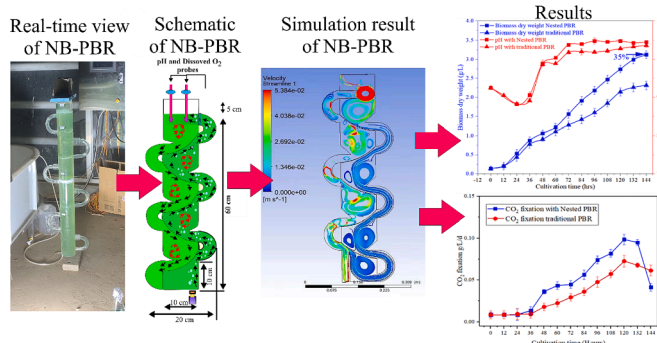
^c School of Agricultural Engineering, Jiangsu University, Zhenjiang 212013, China

^d Qilu University of Technology (Shandong Academy of Sciences), Shandong Analysis and Test Centre, Jinan 250014, China

HIGHLIGHTS

- Nested-bottled photobioreactor (NB-PBR) enhanced mixing through vortex flow field.
- NB-PBR structure increased mixing time increased mass-transfer by 13.33% and 42.9%.
- The overall photochemical efficiency of *A. platensis* was significantly increased.
- Sufficient availability of NaHCO₃ led to increased CO₂ fixation by 35.9%.
- *A. platensis* increased biomass production by 35 % and reached to 3.12 g/L.

GRAPHICAL ABSTRACT



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ABSTRACT

In this study, computational fluid dynamics were employed to examine clockwise and anticlockwise vortices in the rising and down coming sections of novel nested-bottle photobioreactor. The radial velocity was increased by four times which significantly reduced dead zones compared to traditional PBR. The (NB-PBR) comprised of integrated bottles connected by curved tubes ($d = 4$ cm) that generated dominant vortices as the microalgae solution flows through each section ($h = 10$ cm). The (NB-PBR) was independent of the inner and outer sections which increased the mixing time and mass-transfer coefficient by 13.33 % and 42.9 %, respectively. Furthermore, the results indicated that the (NB-PBR) showed higher photosynthesis efficiency preventing self-shading and photo-inhibition, resulting in an increase in biomass yield and carbon dioxide fixation by 35 % and 35.9 %, respectively.

* Corresponding author.

E-mail addresses: ameerali.kubar@yahoo.com (A.A. Kubar), santosh_malhi@live.com (S. Kumar), liuweics@qlu.edu.cn (W. Liu), shjzhcuiyi@hotmail.com (Y. Cui), feifzhu@ujs.edu.cn (F. Zhu), xuxiangru@ujs.edu.cn (X. Xu), 1043995168@qq.com (C. Shao), 1402530010@qq.com (X. Hu), ncookie66@hotmail.com (N.Y.A. Prempeh), huo@ujs.edu.cn (S. Huo).

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1. Introduction

Fossil fuel consumption, food, wastewater generation, and deforestation have significantly increased in recent decades to cope with the increasing demands of global populations (Sung et al., 2022). It is essential to develop sustainable ways to overcome the current energy and food crisis globally (Banerjee et al., 2019). Significant efforts are needed to enhance carbon capture and utilization (CCU) technology to reduce atmospheric carbon dioxide (CO₂) emissions, primarily from power plants and various industrial activities (Song et al., 2019). CCU can reduce carbon emissions from carbon-intensive sectors while producing beneficial byproducts (Pires, 2019).

As the world progresses toward a more sustainable and bio-based economy, microalgae have proven to be the most effective CCU technology, as it is a photosynthetic organism grown on non-arable land using solar energy to transform inorganic carbon molecules into biomass and offer a sustainable approach to achieve carbon neutrality. Microalgae can be processed into numerous products, for example, in biofuel, chemicals, pharmaceutical, food, and feed industries (Raeisossadati et al., 2019), bioplastics, and other valuable products, such as β -carotene (Sui et al., 2019), astaxanthin (Cheng et al., 2016), and omega-3 fatty acids (Cheng et al., 2020b; Daneshvar et al., 2022). Microalgae have garnered significant attention from the food, feed, and pharmaceutical industries due to their large market size. Wastewater treatment of various industries cultivating microalgae is also a well-established potential avenue because microalgae possess a strong ability to treat wastewaters (Hu et al., 2021; Liu et al., 2021).

Large-scale microalgal production facilities have been developed globally to benefit from microalgae's enormous potential (Kumar et al., 2015). Microalgal cultivation systems are typically classified as open ponds and photobioreactors (PBRs). Large-scale farming is frequently considered an open system due to its inexpensive construction and ease of operation (Cheng et al., 2015a). However, PBRs allow for increased exact control of operating conditions while providing a larger biomass output. The enclosed container of the photobioreactor reduces contamination and ensures high-quality biomass, low power consumption, sufficient mixing, and efficient mass transfer (Touloupakis et al., 2022).

Since 2000, PBRs have been recognized as the most promising devices for acquiring bulk biomass, and several innovative PBRs have been developed and refined (Wang et al., 2021). Mixing, gas-liquid mass transfer, and light usage are critical in designing PBRs. Increased gas-liquid interface duration or the interfacial area involving CO₂ bubbles and the solution promote mass transfer. Jingwei Fu et al. analyzed the impact of static airfoil-shaped deflectors in a traditional column photobioreactor. The simulation revealed that the addition of these deflectors resulted in an 11.5 % increase in the mass transfer coefficient and a 21.4 % decrease in mixing time. The enhanced hydrodynamic flow was found to boost the maximum algal biomass yield by 18.3 % and the CO₂ fixation rate by 10.9 % (Fu et al., 2021). Many studies have been conducted on enhancing the above attributes to achieve higher biomass productivity. In serial-lantern constructed PBRs, clockwise and counterclockwise vortices were created in the outer down-flow zone, causing disturbances in algal solutions' radial and axial velocities and turbulent kinetic energy, enhancing biomass and CO₂ fixing by 30 % and 35 % (Ye et al., 2018b). Furthermore, Qing ye et al. developed an innovative tangential spiral-flow vertical PBR that increased biomass output by 1.40 times and CO₂ fixation by 53 %. (Ye et al., 2020). Researchers developed a helical baffle to create a spiraling ascent of CO₂ bubbles, which increased microalgal biomass growth by 40.8 %. (Xu et al., 2020). Novel horizontal tubes and triangular prisms improved biomass by 90 % (Yang et al., 2016). Tesla-valve baffles in column PBRs generated clockwise and anticlockwise vortices and improved biomass yield by 28.1 % (Kubar et al., 2021). Ao Xia et al. installed perforated inverted arcs, airfoil-shaped deflectors, triangular prisms, and inclined baffles to promote biomass yield by 20.9 %, 18.3 %, 70 %, and 25.8 %, respectively (Xia et al., 2018). The primary benefit of PBRs must include a large, illuminated surface area/volume, uncomplicated construction, efficient mixing, easy maintenance and less constructing cost. These baffles help mixing, light absorption, and mass transfer efficiency by incorporating inner static mixers (or baffles) and subsequent structural optimizations (Chen et al., 2016; Yang et al., 2016).

Nevertheless, the airlift column PBR with a draft tube has certain drawbacks. The traditional PBR has two sections, inner and outer (riser and downcomer), separated by a baffle. The riser (inner) section only receives proper or enough light during the initial cultivation stage, but as the microalgae become dense, it is challenging for light to penetrate to the PBR's central zones, resulting in poor microalgae photosynthesis, as long as microalgae stays in the inner section. The light intensity gradually decreases as the distance from the illuminated side of the photobioreactor increases, limiting microalgal cell proliferation.

In this study, a unique nested-bottle photobioreactor (NB-PBR) was designed and developed with the aim of increasing mass transfer by extending the time that gas bubbles come into contact with the liquid. The NB-PBR design mainly modified the inner and outer sections (riser and downcomer) commonly found in traditional airlift column photobioreactors, which can limit the growth of *A. platensis* by reducing light availability. The dominant vortex flow created in each part of the NB-PBR enhanced mixing and mass transfer. In addition, a computational fluid dynamics model was used to confirm the overall mixing performance. The optimal NB-PBR structure was determined by altering various structural dimensions. Finally, the actual photochemical efficiency was examined in order to achieve higher microalgal growth. The NB-PBR also accelerated the algae movement into light/dark zones and reduced self-shading, nurturing a friendly environment to accumulate higher algae yield.

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2. Material and methods

2.1. Numerical simulation for the vortex flow field in the NB-PBR

A three-dimensional mesh of the photobioreactor was created using the Ansys ICEM CFD software (64-bit, Ansys, USA). Ansys Fluent 2021 R1 was then used to perform a computational fluid dynamics simulation. The gas outlet included a degassing barrier, which prevented the continuous phase gas from passing through the top surface and allowed it to escape only in the dispersed phase Huang et al., (2015b). The numerical model's boundary conditions were as follows. The gas outlet was configured to the degassing boundary, and the bioreactor walls and both channels were set to no-slip wall conditions. The gas-liquid flow was simulated using a homogeneous Volume of Fluid (VOF) model. Grid independence was assessed using 210,000 and 310,000 cells, and the results changed a little. The time-step size was 0.005 s for 10,000 time-steps. The overall velocity contours of the NB-PBR were recorded, and the computation was performed twice consecutively on separate machines.

2.2. Designing NB-PBR configurations

Fig. 1 shows a schematic of the NB-PBR. The NB-PBR was entirely redesigned to generate more vortices in the solution to promote the performance of vertical traditional column PBRs. The NB-PBR comprised two sections: the main central column and curved arms on both sides were attached to the central column. The main central column was $d = 10$ cm and $h = 60$ cm, whereas the curved-arm diameter was changed to 2, 2.5, 3, 3.5, and 4 cm. The NB-PBR had riser and downcomer portions, each containing three regions with a height of 10 cm. Aeration was provided from the bottom right. The uniquely designed NB-PBR altered the traditional design of vertical column reactors with high efficiency because it provided enough light to strengthen photosynthetic efficiency in *A. platensis* during rising and down-coming flows.

Meanwhile, the traditional PBR consisted of two sections inner (riser) and outer (downcomer) section with diameter ($d = 10$ and 20 cm). The standard growth conditions include light ($12,000 \pm 200$ lx), a temperature of 27.2°C , with 15% CO_2 aeration (0.1 vvm).

2.3. Measurement of mixing time (MT) and mass transfer coefficient (MTC)

A pH probe (InPro3253i/SG/120 Mettler Toledo) was used to assess mixing time (MT). The NB-PBR was filled with 7 L of clean water at first, then the pH was adjusted to 3 by adding 35% HCL. An alkalinity trace was obtained after addition of sodium (2.5 – 3.5 mL of 12 mol/L), and the pH probe displayed two consecutive peaks. The MT was calculated by subtracting these two consecutive measurements (initial from the final). The mass transfer coefficient (MTC) was then determined by adding N_2 gas to the solution, which reduced the O_2 concentration to 4 mg/L, followed by the addition of air to increase the concentration of dissolved oxygen to 5 mg/L. An oxygen probe (Mettler Toledo InPro6850i/12/120) was utilized to automatically save data every 0.1 s. To connect the two probes, a transmitter (i-7017fc, ICP DAS, Taiwan) was utilized, and the final data was obtained using software. (Cheng et al., 2015b).

2.4. Measurement of actual photochemical efficiency

The OJIP test, which involves the measurement of the rapid induction kinetic curve of chlorophyll fluorescence, is often used to study the photochemical reaction of photosystem II (PSII) in microalgal cells after they have been kept in darkness for at least 20 min. This test is performed using the OJIP method. This investigation used AquaPen-PAP 110-P to record the OJIP curve. Multiple parameters were gathered based on OJIP curves. For instance, the PSII quantum yield, the highest quantum efficiency of PSII, is calculated as follows. $F_v/F_m = (F_m - F_o)/F_m$. In addition, specific activity indicators including ABS/RC and DIO /

RC, which represent the absorbed and dissipated light energy of the unit PSII response center, were evaluated. The indicators TRo/RC and ETo/RC indicate the light energy used in the unit PSII reaction center for QA reduction and electron transport, respectively.

2.5. Measurement of total inorganic carbon (TIC), sodium carbonate (Na_2CO_3), and sodium bicarbonate (NaHCO_3)

The double-tracer approach was used to determine the concentration of inorganic carbon in the solution. A 10 mL sample of the solution was filtered and transferred to a conical flask, where one drop of 1% phenolphthalein was added before gently adding 0.1 mol/L H_2SO_4 to the flask while stirring until the sample became colourless. The sample was then titrated again with 0.1 mol/L H_2SO_4 after adding two drops of 0.1% methyl-orange until the colour changed from yellow to orange (Cheng et al., 2018a). The concentrations of Na_2CO_3 and NaHCO_3 were calculated using the quantity of H_2SO_4 utilized and the pH of the solution. To assure accuracy, each test was performed three times, and the average and standard deviations were recorded.

2.6. *A. platensis* cultivation and biomass dry weight measurement

Zarrouk's medium was used to culture a mutant *Spirulina* strain of *A. platensis* acquired ZJU9000 from Zhejiang University (Cheng et al., 2017). A vacuum pump harvested the dry biomass mass (Model: DOA-P504A-BN, USA). To measure the weight of the sample, dry filter paper was weighted and then 10 mL sample was filtered on paper using a vacuum, and dried in an oven at 50°C for three hours. The initial weight of the dry filter paper was subtracted from the final weight to obtain the weight of the sample.

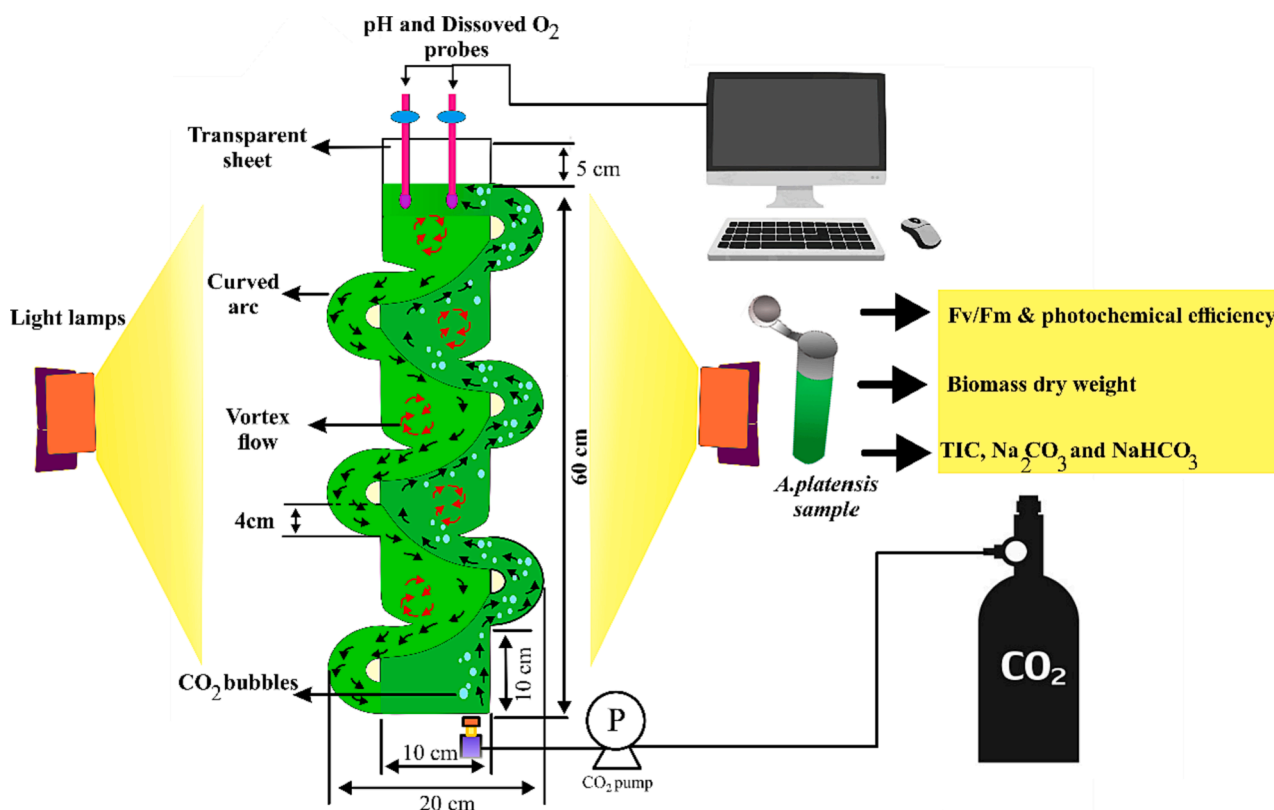


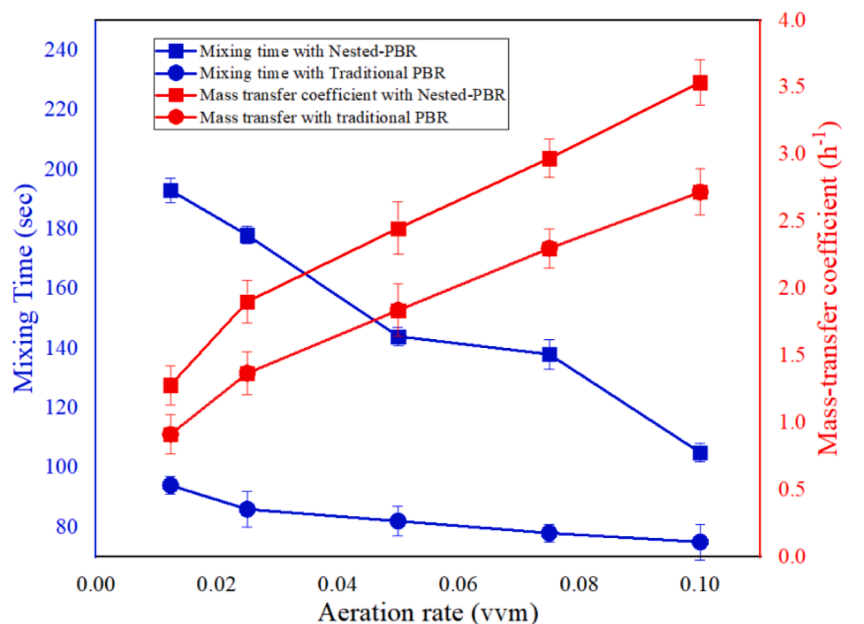
Fig. 1. A schematic of the nested-bottle (NB) photobioreactor (PBR) (green: algae solution, red: vortex flow fields, circles: CO_2 bubbles, orange: sparger). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Results and discussion

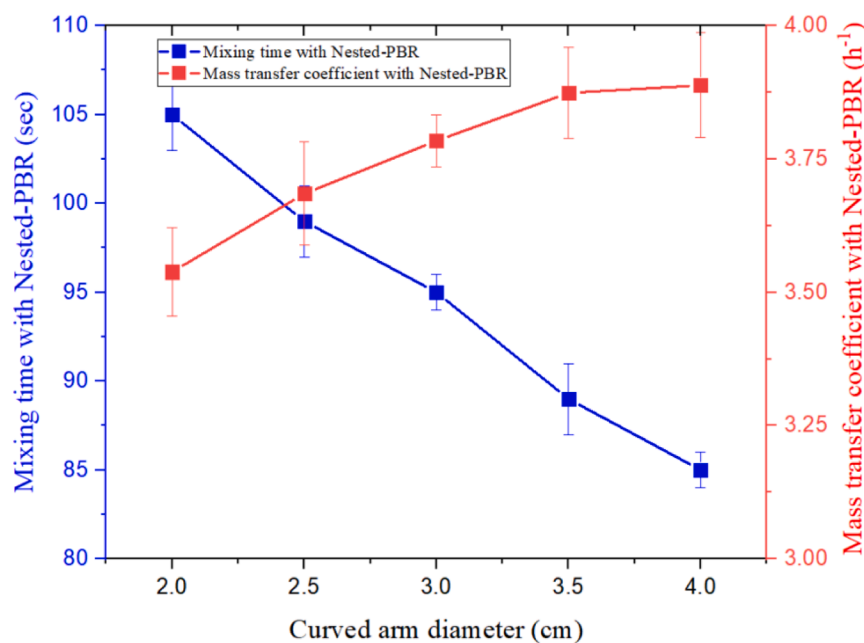
3.1. The NB-PBR's increased MT and MTC

The MTC indicates the PBR's mass transfer capacity, whereas the MT represents the PBR's mixing capacity. Under normal conditions, CO_2 is a limiting factor for microalgal growth, and as the mass transfer capacity is improved, more CO_2 is dissolved (Pegallapati and Nirmalakhandan, 2013). Consequently, the PBR's mass transfer capacity must be improved, and structural factors heavily influence the total performance

of an efficient column PBR. Recent research indicates that incorporating baffles in (PBRs) and raceway ponds typically results in the formation of a vortex flow fields and has been proven beneficial in achieving a higher biomass yield of microalgae (Cheng et al., 2015b, 2018c; Kumar et al., 2022). The NB-PBR caused a periodic change in the straight flow cross-section area after passing through curved arm on the side of the main central column. Strong clockwise and anticlockwise vortices were formed in the regions, generating an enhanced radial flow field (Ye et al., 2018c). Mixing the cells causes them to move, increasing the surface area exposed to light, maintaining a stable solution temperature,



a.



b.

Fig. 2. The NB-PBR mixing time (MT) and mass transfer coefficient (MTC). (a) Effect of aeration on MT and MTC (b) Effect of curved-arm diameter on MT and MTC.

and improving the rate of CO₂ fixation through flow vortices formed by the structure (Cheng et al., 2019a; Ye et al., 2018a).

Several approaches have been suggested to enhance the mass transfer capacity (MTC) in the production of microalgal biomass. For example, in a flat plate PBR, mechanical agitation can be used to create additional turbulence and shear, which can break large bubbles into smaller ones and increase the MTC by 41.2 % (Cheng et al., 2018d).

The only method to disrupt the flow in both PBRs was through aerating the CO₂ bubbles from the bottom. CO₂ was continuously bubbled through the water, where it dissolved and provided a source of carbon for microalgae to grow. Consequently, the longer the bubble travelled in the solution, the better it was for CO₂ dissolution and led to rapid interaction of CO₂ and water. Results showed, increasing the aeration rate shortened the MT and enhanced mass transfer. When the gas aeration rate was increased from 0.0123 to 0.1 vvm with a curved-

arm diameter of 2 cm Fig. 2a, the MT decreased from 193 to 105 sec and the MTC increased from 1.28 to 3.53 h⁻¹ in the NB-PBR respectively. Fast-moving bubbles significantly boosted the turbulence intensity, resulting in a quick dissolution rate and improved mass transfer performance. The MT and MTC were also optimized Fig. 2b for cultivation by varying the curved-arm diameter (2, 2.5, 3, 3.5, and 4 cm), whereas the aeration was kept constant at 0.1 vvm. The results showed that the MT further decreased from 105 to 85sec, and the MTC increased from 3.53 to 3.88 h⁻¹ in the NB-PBR when the curved-arm diameter reached at 4 cm. It was concluded that the increase in the curved-arm diameter allowed more fluid to pass through and rise because a wider tube allowed sufficient space than a narrow tube, increasing the mixing and decreasing the time spent. Improved mixing of nutrients and CO₂ dissolution in a PBR significantly enhanced the conversion of HCO₃⁻ ions consumed during efficient production of *A. platensis* (CO₂ + H₂O →

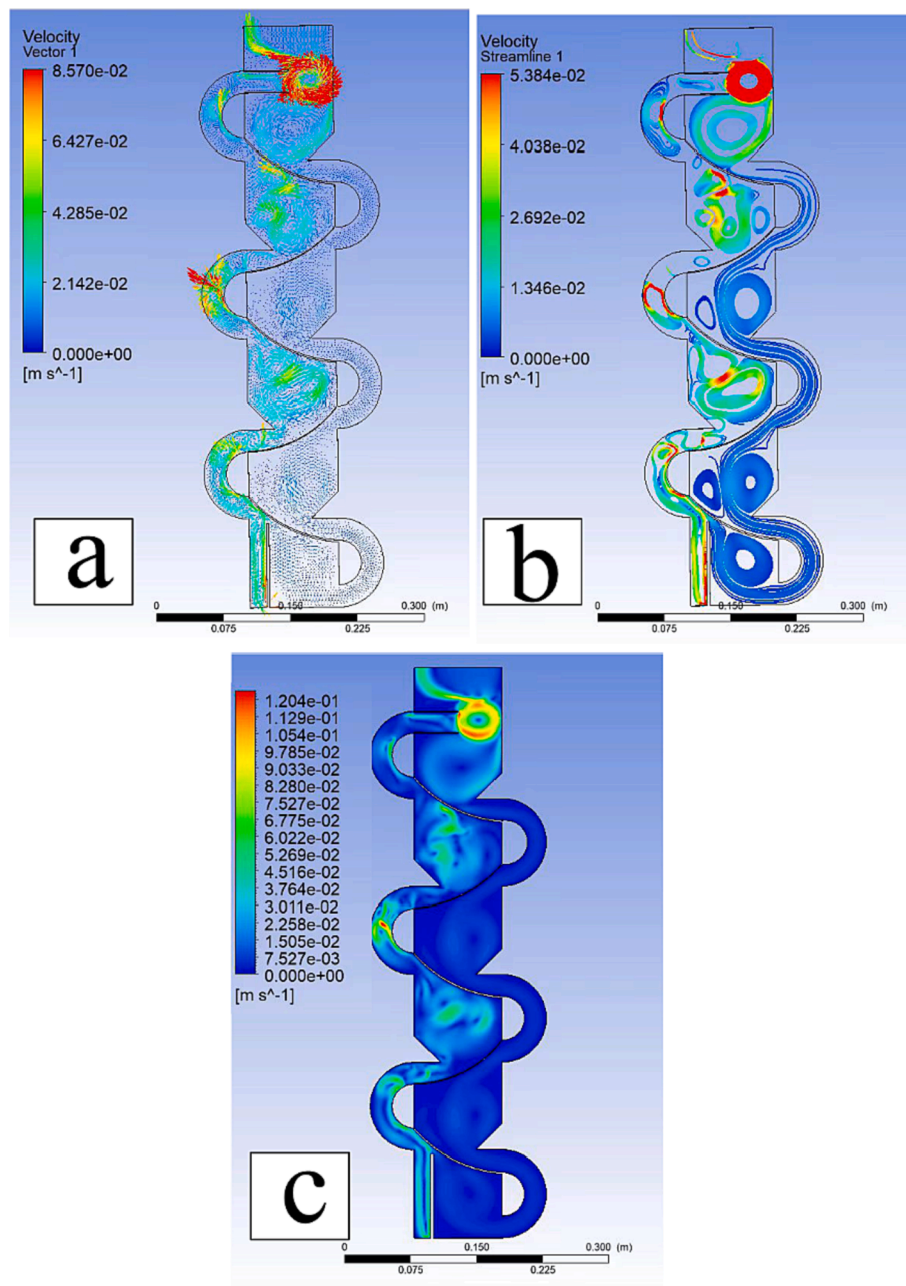


Fig. 3. Velocity magnitude contours and vector diagrams of the microalgal solution in the NB-PBR. (a) Velocity vector diagram (b) Velocity magnitude contour (c) Velocity streamlines.

$\text{HCO}_3^- + \text{H}^+$).

The unique shape of the NB-PBR provided sufficient resistance to CO_2 bubbles while rising, which otherwise goes unutilized from the algae solution as an idle resource. The curved arms attached to both sides of the NB-PBR forced the CO_2 bubbles to remain in the solution longer and resist easy escape due to the formation of vortices. It improved nutrient dissolution and diffusion by increasing the gas dissolution rate and mass transfer efficiency. The primary novelty of the NB-PBR was that several vortices were observed in the riser and downcomer sections, which significantly improved the overall performance of the column PBR.

3.2. Liquid flow distribution in NB-PBR with simulation

The NB-PBR created vertical clockwise and anticlockwise vortex flow fields each time the solution flow passed the curved arms and entered the main column sections. Most researchers have made several alterations in the traditional column PBR to generate vortices (Cheng et al., 2018b; Song et al., 2021; Ye et al., 2018a). The NB-PBR's entire flow field was divided by a riser coupled with a down-flow sections, but no inner or outer areas existed, considerably different from typical PBRs, which are typically constructed with inner and outer flow regions for circulating microalgae solutions. Consequently, an inefficient process of mass transfer and light exposure occurs. The main reason is that, the straight central tube is inadequately supplied with light and smooth structure hardly offers any disturbance in the flow. However, the NB-PBR with an innovative structure completely disturbed the flow patterns, forming clockwise and anticlockwise vortices in both regions. Fig. 3a shows the velocity vectors, showing that the solution's velocity was improved in the riser section and with a high-velocity magnitude Fig. 3b, whereas Fig. 3c shows clear vortices while rising and downcoming.

The primary vortex's higher energy and velocity permitted efficient dissolving and mixing of CO_2 and nutrients and prevented microalgae from sinking or settling. Mixing also prevents cell sedimentation on the reactor's bottom and walls and the removal of O_2 from the culture (Amini et al., 2016). The inclusion of secondary vortices in a narrow-bodied photobioreactor (NB-PBR) improved the mixing of nutrients and dissolved CO_2 , as well as the movement of microalgae between light and dark zones, which is beneficial for photosynthesis. These vortices also enhanced the mass transfer and flashing light effects in the PBR, leading to increased microalgae growth (Fu et al., 2021; Ye et al., 2018b).

Fig. 3 shows the simulation results of the NB-PBR's flow fields. Simulating the cultivating conditions in the NB-PBR confirmed the velocity magnitudes and vortices. The movement of the solution flow was only maintained by aerating CO_2 bubbles from the bottom, which lifted the solution up while flowing from the curved arms and the main section. The simulation results showed that vortices were generated in the PBR's curved arms and main column regions, and the rising and downcoming sections are identical in the NB-PBR, supporting the formation of vortices in each section.

The velocity reached 0.04–0.05 m/s, whereas the radial velocity in the traditional PBR's solution was low at 0.01 m/s when the input gas rate was 0.12 vvm. Hence, the NB-PBR improved the radial velocity by four times. Consequently, the greater radial flow facilitated microalgae to move frequently between dark and light sections, hence improving the flashing-light effect and significantly influencing microalgal photosynthesis and biomass growth. The six uniquely designed sections (three each in the riser and downcomer) were designed to generate a widely distributed flow regime, providing the PBR's microalgae at different depths with a good light/dark cycle period. The visible simulated vortex flow field prevented the accumulation of oxygen generated by photosynthesis, reducing the occurrence of dead zones (Aslanbay Guler et al., 2019).

3.3. NB-PBR increased the actual photochemical efficiency and OJIP values

As shown in Fig. 3, the unique design of the NB-PBR resulted in a vortex flow that enabled microalgae cells to circulate between the light and dark zones, improving their photosynthetic efficiency and contributing to a high growth rate. In contrast, if the microalgae culture lacks an effective velocity flow and mixing, many cells may not receive sufficient light exposure and a regular light/dark cycle for photosynthesis, leading to reduced biomass development. The NB-PBR also raised the microalgal solution's radial velocity, enhanced the microalgae's flashing effect, and elevated the photosynthetic efficiency.

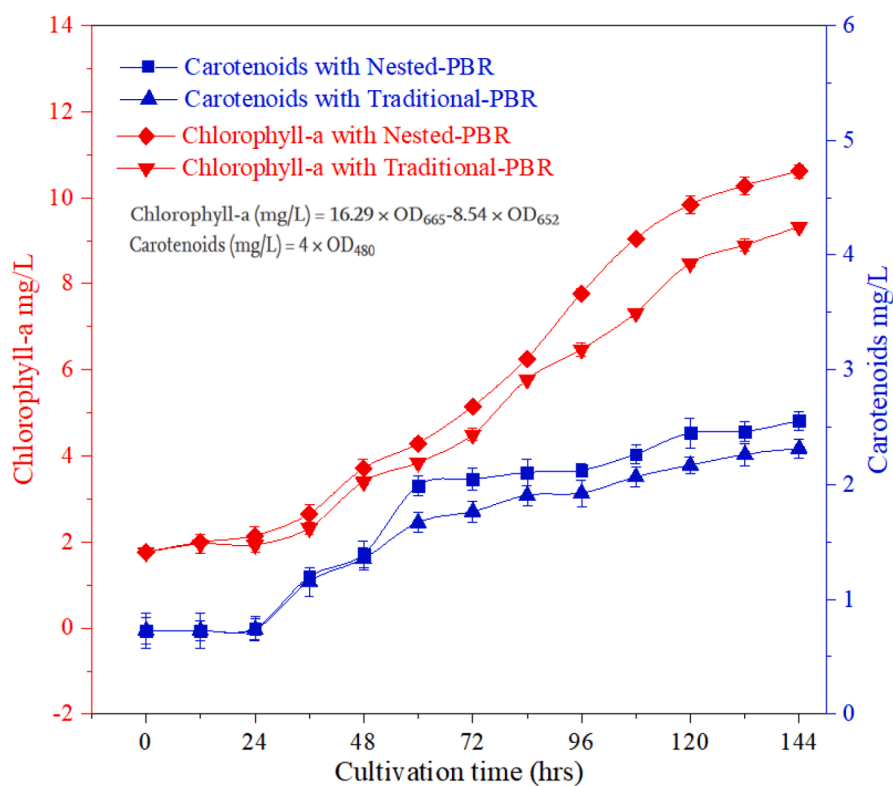
Over 60 % of chlorophyll is located in the light-capturing antenna complex. This complex turns light energy into excitement for the photosynthetic center. Carotenoids help antenna systems and chlorophyll-binding proteins gather light and reduce free radicals in microalgal cells. Higher chlorophyll-a content in *A. platensis* cells increases photosynthetic efficiency. The amount of chlorophyll-a and carotenoids in cells in the NB-PBR was on average 10.61 mg/L and 2.55 mg/L, respectively, and these amounts increased by 13 % and 10 % respectively Fig. 4. Higher chlorophyll-a and carotenoids in NB-PBR cells increased *A. platensis* photosynthesis and CO_2 fixation.

The chlorophyll fluorescence (OJIP) was employed to investigate the photosynthetic activity in *A. platensis* influenced by the structural changes in the NB-PBR and traditional PBR (Fig. 5). Samples from both PBRs were dark adaption for 15 min, and OJIP curves were obtained using 2 s pulses of red light. Three channels absorb energy; it's employed for photochemical reactions (photochemistry), fluorescence, or heat. Chlorophyll fluorescence parameter analyses provide significant data regarding energy absorption, use, transmission, and electron transfer status in the photosynthetic electron transport chain.

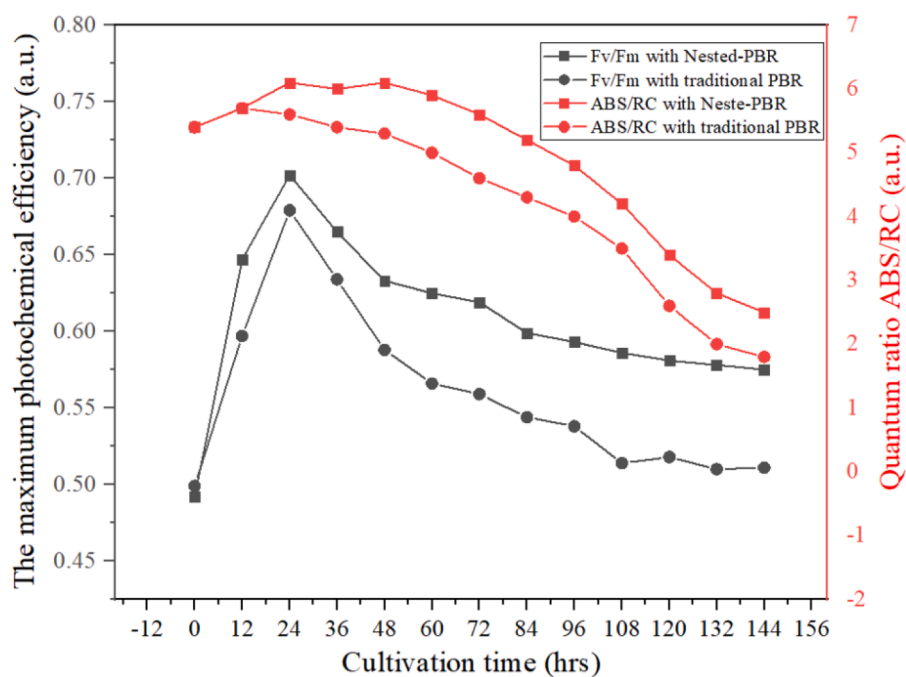
As the most frequent fluorescence parameter, Fv/Fm can indicate the PSII system's activity and is one of the most measured photosynthetic parameters to characterize PSII reaction conversion and photosynthetic efficiency. The Fv/Fm and ABS/RC values were higher in NB-PBR Fig. 4 due to the lower density of *A. platensis* in the solution, allowing easy light penetration to reach each section of the PBR. Therefore, the maximum photochemical efficiency and the absorbed light energy of the unit reaction center were increased, which increased the possibility of the algal cells accepting more light photons (Gan et al., 2019; Undurraga et al., 2016). However, photosynthetic efficiency decreased with cultivation time as biomass growth increased. The enhanced mixing and mass transfer in NB-PBR improved the overall performance of the PBR, particularly by creating vortex flow fields. The percentage of algal cells that were light-deprived or photodamaged reduced dramatically. Because Fv/Fm is strongly related to light consumption efficiency, the algal cells in NB-PBR were more equally distributed and avoided sedimentation of algal solution, enhancing the biomass yield per photon absorbed (Kumar et al., 2021).

The ABS/RC increased from 1.8 to 2.5, respectively Fig. 4b. The adequate amount of HCO_3^- stimulated the Calvin cycle activity, which increased the consumption of ATP and NADPH, so encouraging the antenna pigment to absorb additional light energy (Cheng et al., 2020a). Moreover, the ETo/RC and TRo/RC increased in the NB-PBR from 0.78 to 0.94 and 2.18 to 2.38, respectively, indicating that the electron transfer chain was accelerated Fig. 5b. The enhanced light energy absorption raised the maximum light energy conversion efficiency of the PSII reaction center (Guo et al., 2019; Li et al., 2020). Furthermore, Dio/RC dropped from 3.6 to 3.31 with NB-PBR, showing less light was utilized for heat dissipation. The antenna pigment absorbs light energy for heat dissipation, fluorescence, and energy conservation (Cho et al., 2019). The lower heat dissipation indicates that a greater proportion of the solar energy was used for photosynthesis (Mao et al., 2021).

The NB-PBR comprised two separate sections, riser and downcomer. The structure was designed to allow maximum light to penetrate the PBR by maintaining homogeneity across the culture. The suggested NB-PBR

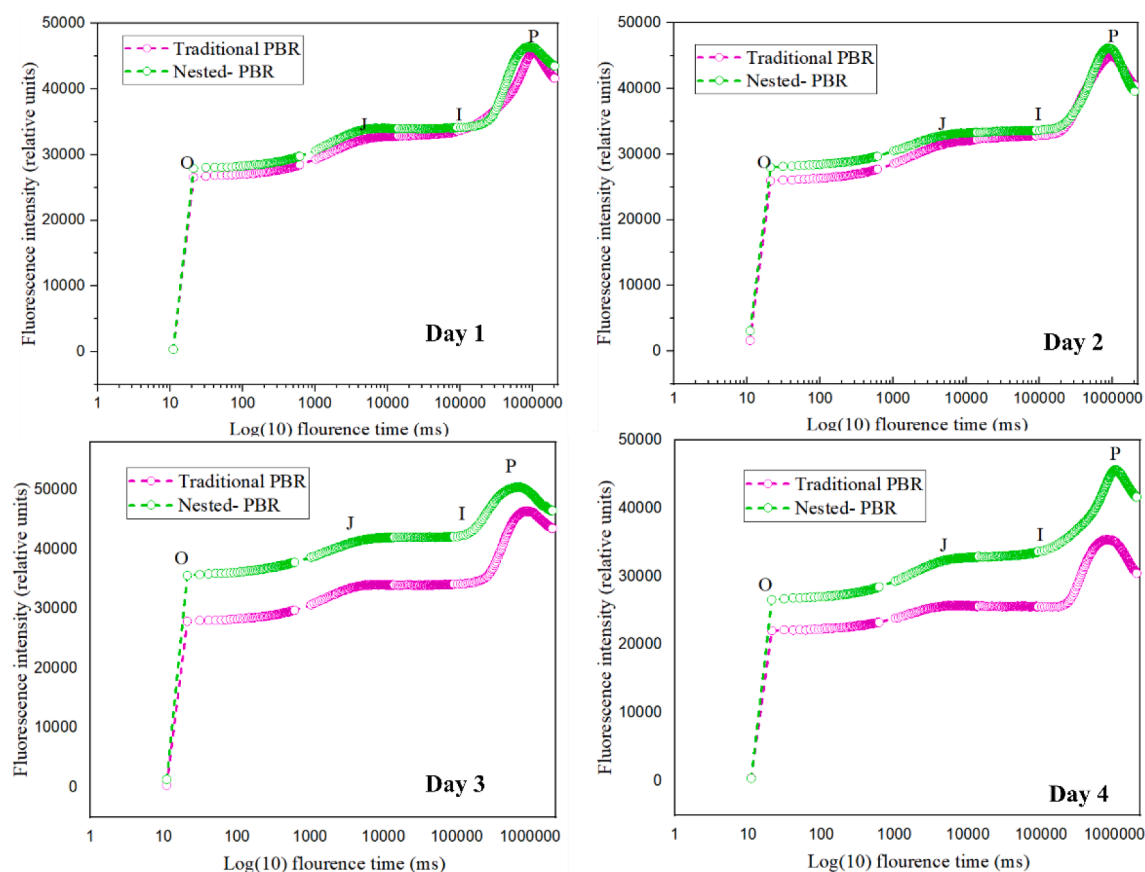


(a) Chlorophyll-a and Carotenoids

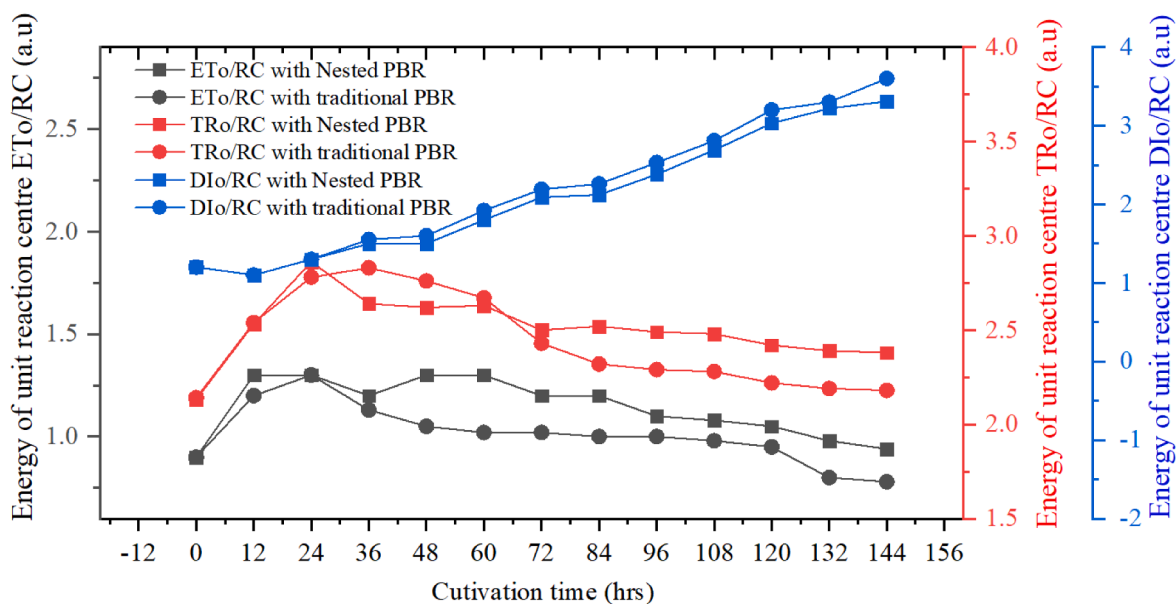


(b). Fv/Fm and ABS/RC

Fig. 4. Comparison of chlorophyll fluorescence induction kinetic parameters of *A. platensis* cells in the traditional photobioreactor (PBR) and nested-bottle (NB)-PBR. (a) Chlorophyll-a and carotenoids (b) Fv/Fm and ABS/RC.



a



b

Fig. 5. (a) OJIP curves and (b) the consumed (DIo/RC) light energy of the unit reaction center and the light energy used for Q_A reduction (TRo/RC) and electron transfer (ETo/RC) of the unit reaction center of microalgal cells in both photobioreactors (PBRs) with different cultivation times.

boosted the microalgae cells' capacity to absorb more light photons; nevertheless, lengthy durations of stay in either area (light or dark) were avoided since it negatively impacts photoinhibition and respiration, affecting *A. platensis* growth. NB-PBR increased energy supply and NADH oxidation, enhancing transmembrane protein transport, ATP synthase, and proton gradient formation. (Kazbar et al., 2019; Zhi et al., 2019).

3.4. NB-PBR promoted the dry biomass and TIC in the solution

A. platensis was cultivated in the NB-PBR under optimized conditions. The microalgal growth rates in both PBRs were similar during the first 24 h, as microalgae's density was low hence both PBRs allowed sufficient light to reach each section of the PBR, which was favorable for microalgae growth. However, after 120 h, the NB-PBR showed a significant increase in biomass production by 35 % (3.12 g/L) relative to the traditional PBR Fig. 6. The vortex fields were absent in the traditional PBR, and as the number of microalgae cells increased, the light exposure and light/dark cycle were negatively affected, reducing photosynthesis and microalgal biomass growth.

Subsequently, the well-designed NB-PBR structure formed vortices across the PBR, uniformly supporting the distribution of nutrients, avoiding sedimentation and stagnation, increasing the gas-liquid contact time, and improving CO₂ dissolution. Rising CO₂ bubbles circulated the liquid flow in both PBRs, and each section in the NB-PBR retained the rising bubbles for a short period before escaping into the next section, breaking the larger bubbles into smaller sizes and improving the gas-liquid mass transfer. Smaller bubbles with a longer retention period enhanced CO₂ mass transfer and hydration, increasing dissolved TIC. Microbubbles could absorb CO₂ better than larger ones.

Typically, microalgal cells absorb CO₂ in two ways. 1) Dissolved CO₂ molecules pass the thylakoid membrane through carbonic anhydrase. 2) Extracellular CO₂ and H₂O react to form HCO₃⁻, which is transferred to the nucleoid and converted to CO₂ during Calvin cycle. In addition to enhancing CO₂ transport through microalgal cells, utilizing HCO₃⁻ as a medium for boosting CO₂ fixation could also be a feasible solution.

Furthermore, the pH value initially decreased quickly during the first 24 h because the *A. platensis* density was lower at that stage. HCO₃⁻ was the result of the chemical reaction of CO₂ and H₂O, while a continuous supply of CO₂ produced excess H⁺ ions. Moreover, NaHCO₃ ions resulted from Na₂CO₃ in the solution (CO₂ in H₂O → HCO₃⁻ → ⁺) and (CO₂ + Na₂CO₃ + H₂O → 2NaHCO₃, HCO₃⁻ → CO₂ + OH⁻) (Cheng et al., 2019b, 2018c).

Thus, the NB-PBR increased the availability of NaHCO₃ ions, which were utilized during the Calvin cycle via photoreaction (Cao et al., 2016). As the microalgae culture grew in both photobioreactors (PBRs), the solution concentration increased, necessitating a higher supply of dissolved inorganic carbon and properly blended nutrients to sustain its fast growth rate. Finally, the NB-PBR accumulated more biomass than the traditional PBR (3.12 g/L vs 2.31 g/L, respectively) because of the increased MTC, improved mixing, and increased photosynthesis. After 24 h, as the *A. platensis* culture grew denser, it gradually used the abundant H⁺ ions via active transport while releasing OH⁻ ions into the solution, increasing the pH value. In contrast to the traditional PBR, the NB-PBR exhibited a higher pH value, consistent with a greater CO₂ fixation by *A. platensis* Fig. 6c.

The NB-PBR significantly generated vortices with its uniquely well-thought structure, which led to adequate mixing across the entire PBR and positively affected the CO₂ fixation. Furthermore, as the microalgae concentration increased, the NB-PBR resulted in a uniform dispersion of algal cells between the light and dark zones more frequently, enhancing photosynthetic efficiency and increasing the overall biomass yield.

4. Conclusions

The CFD simulation using (NB-PBR) examined dominant clockwise

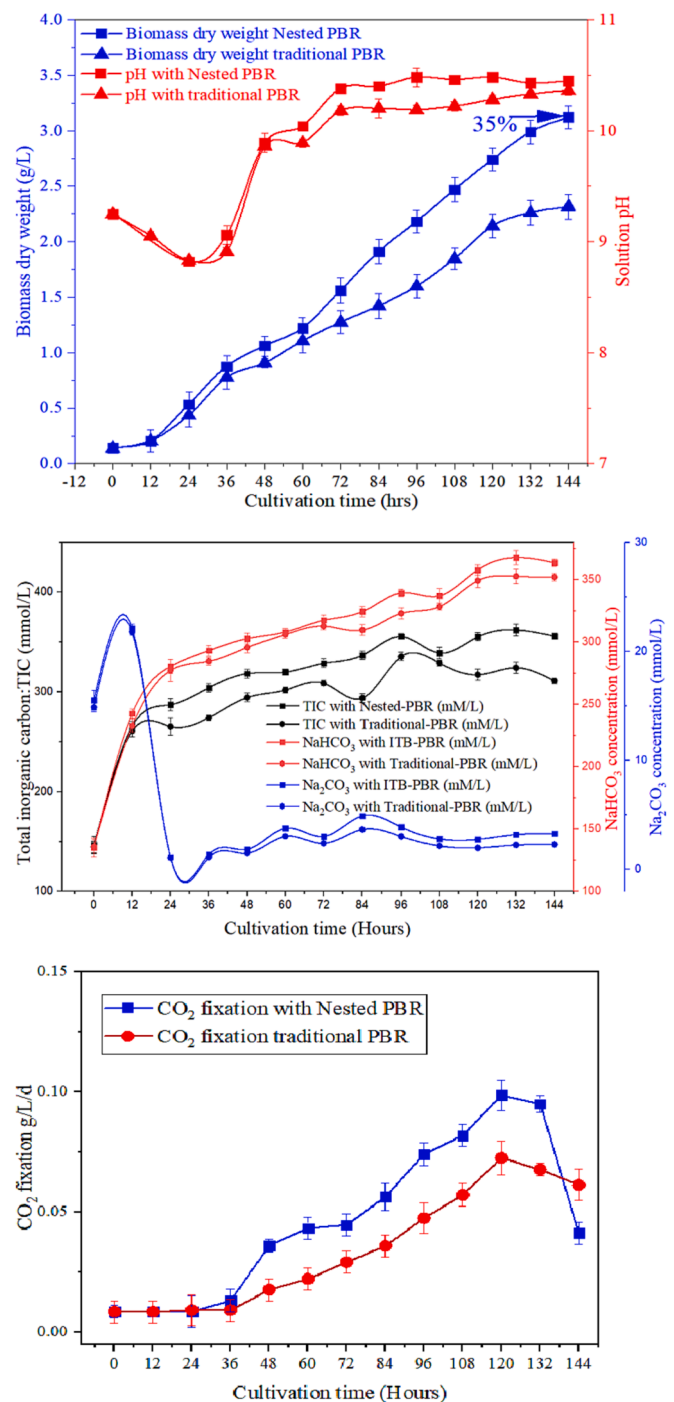


Fig. 6. (a) *A. platensis* biomass dry weight and pH value with 15% CO₂ (where “p” is the carbon content in microalgae dry weight (50%)) (b) The total inorganic carbon (TIC), Na₂CO₃, and NaHCO₃ concentrations in the *A. platensis* solution in the nested-bottle (NB)- photobioreactor (PBR) and traditional PBR (c) CO₂ fixation.

and anticlockwise vortex flow patterns in both the riser and down coming sections and increase biomass yield by 35 %. The NB-PBR minimized dead zones and provided better light availability in each section. Additionally, the unique structure of the NB-PBR promoted mixing throughout the flow, leading to increased mass transfer and CO₂ fixation rates by 42.9 % and 35.9 % respectively. These significant results justified the NB-PBR as a potentially viable option for *Arthrospira platensis* cultivation and efficient CO₂ fixation.

CRediT authorship contribution statement

Ameer Ali Kubar: Conceptualization, Methodology, Data curation, Writing - original draft. **Santosh Kumar:** Carried out the experiments, Formal analysis, Investigation. **Wei Liu:** Formal analysis, Investigation. **Yi Cui:** Methodology, Formal analysis, Investigation. **Feifei Zhu:** Formal analysis, Investigation. **Xiangru Xu:** Formal analysis, Investigation. **Cong Shao:** Carried out the experiments, Formal analysis, Investigation. **Xinjuan Hu:** Methodology, Formal analysis, Investigation. **Nama Yaa Akyea Prempeh:** Carried out the experiments, Formal analysis, Investigation. **Shuhao Huo:** Conceptualization, Resources, Writing - review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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